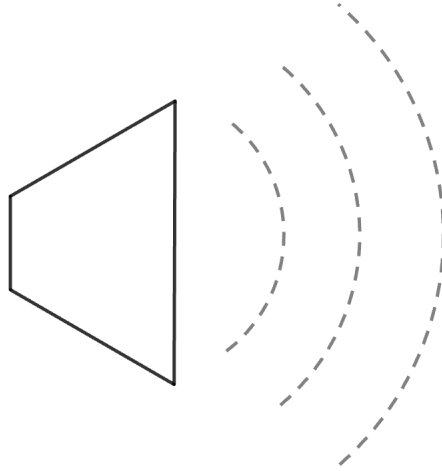


2012 F=ma Exam: Problem 22

Kevin S. Huang



To figure out the base units of power, recall the equation $P = Fv$. Taking the units of both sides,

$$[P] = [F][v] = [F] \frac{\text{m}}{\text{s}}$$

To figure out the base units of force, recall the equation $F = ma$. Taking the units of both sides,

$$[F] = [m][a] = \text{kg} \cdot \frac{\text{m}}{\text{s}^2}$$

Substituting this in the previous equation,

$$[P] = \text{kg} \cdot \frac{\text{m}^2}{\text{s}^3}$$

Since by definition $[P] = \text{W}$, we have

$$\text{W} = \frac{\text{kg} \cdot \text{m}^2}{\text{s}^3}$$

Finally,

$$\frac{\text{W}}{\text{m}^2} = \frac{\text{kg}}{\text{s}^3}$$

so the answer is $\boxed{\text{A}}$.